



SPARKLE: Symbolic caPtuRing of knowledge for Knowledge graph enrichment with LEarning

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ABSTRACT

Knowledge graphs (KGs) naturally capture the convergence of data and knowledge, making them expressive frameworks for describing and integrating heterogeneous data in a coherent and interconnected manner. However, based on the Open World Assumption (OWA), the absence of information within KGs does not indicate falsity or non-existence; it merely reflects incompleteness. Inductive learning over KGs involves predicting new relationships based on existing statements in the KG, using either numerical or symbolic learning models. The Partial Completeness Assumption (PCA) heuristic efficiently guides inductive learning methods for Link Prediction (LP) by refining predictions about absent KG relationships. Nevertheless, numeric techniques—like KG embedding models—alone may fall short in accurately predicting missing information, particularly when it comes to capturing implicit knowledge and complex relationships. We propose a hybrid method named *SPARKLE* that seamlessly integrates symbolic and numerical techniques, leveraging the PCA heuristic to capture implicit knowledge and enrich KGs. We empirically compare *SPARKLE* with state-of-the-art KG embedding and symbolic models, using established benchmarks. Our experimental outcomes underscore the efficacy of this hybrid approach, as it harnesses the strengths of both paradigms. *SPARKLE* is publicly available on GitHub¹.

CCS CONCEPTS

• Information systems → Data management systems.

KEYWORDS

Inductive Learning, Symbolic Learning, Knowledge Graphs

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¹<https://github.com/SDM-TIB/SPARKLE/>



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1 INTRODUCTION

Knowledge graphs (KGs) have garnered significant attention as versatile data structures that effectively merge data and knowledge through factual statements [1, 8, 10]. Both scientific communities and practitioners have played an active role in developing KGs, producing both encyclopedic and domain-specific variants. However, because of the Open World Assumption [13], KGs cannot be assumed to be complete. Inductive learning over KGs encloses various methods to capture knowledge in KGs that will support KG completion. Inductive learning, a crucial process for capturing missing links within KGs, involves deducing patterns and relationships from existing data. Established approaches can learn symbolic or numerical representations of KGs' patterns, which correspond to building blocks for inferring missing links [10]. Thus, effective inductive learning has become vital for KG completion [2, 3, 22].

Problem Statement and Proposed Solution: This paper tackles the challenge of KG completion by introducing a novel hybrid approach. Our method extends the capabilities of numerical Knowledge Graph Embedding (KGE) models by seamlessly integrating insights derived from patterns unveiled through symbolic learning [18, 21]. By doing so, we aim to transcend the current challenges posed by incomplete KGs. Leveraging the Partial Completeness Assumption (PCA) heuristic, our approach not only contributes to enhancing KG completion but also holds the potential to improve the predictive performance of KG embeddings. The proposed method called *SPARKLE*, implements our hybrid approach by effectively guiding the application of the PCA heuristic to facts inferred from Horn rules acquired through symbolic learning. Built on design patterns recently proposed in the literature [19, 20], *SPARKLE* is conceptually elucidated through a *compositional* design pattern that combines symbolic and numerical learning [6, 18, 21, 22].

Empirical Evaluation: We conduct an empirical evaluation of *SPARKLE* where the symbolic system is implemented using the algorithm introduced by Lajus et al. [12]. Simultaneously, we utilize several state-of-the-art KGE models to generate low-dimensional embeddings. Our evaluation strategy involves applying *SPARKLE* to established benchmarks for LP. We quantify the performance of *SPARKLE* using metrics such as Mean Reciprocal Rank (MRR)

and Hits@K. Our study encompasses a total of 144 testbeds. Among these, we utilize 96 testbeds to assess the effectiveness of the *SPaRKLE* approach, while the remaining 48 are employed to determine the computational overhead associated with the hybrid method. Our findings reveal that in 85 out of 96 testbeds, *SPaRKLE* demonstrates superior performance to state-of-the-art alternatives regarding MRR or Hits@K scores. Notably, despite the requirement of computing both symbolic and KGE model components, the execution time overhead introduced by *SPaRKLE* is negligible. These results underscore the advantages of applying the PCA heuristic in enhancing prior knowledge to empower KG completion models.

Contributions: This work contributes as follows: 1) *SPaRKLE*, a hybrid system able to enhance inductive learning models with knowledge captured from Horn rules mined using symbolic learning; 2) Empirical evaluation comparing state-of-the-art KGE and symbolic learning models using existing benchmarks; 3) Empirical studies putting into perspective the benefits of capturing knowledge following the PCA heuristics and their impact on Link Prediction. The rest of the paper is structured as follows: Section 2 defines basic concepts of inductive learning, and motivates our work with an example. Section 3 presents our problem statement and main properties of SPARKLE. Results of the experimental study are reported in Section 4. Section 5 discusses the state of the art. Finally, we close with the conclusions and future work in Section 6.

2 PRELIMINARIES AND MOTIVATION

Knowledge Graph. Given a set *Con* of countable infinite constants. A *knowledge graph* (*KG*) is a directed *edge-labeled graph* $KG = (V, E, L)$, where (1) $V \subseteq Con$ is a set of nodes, (2) $L \subseteq Con$ is a set of edge labels, and (3) $E \subseteq V \times L \times V$ is a set of edges.

The set *E* of edges is also known as set of *positive edges* of *KG*, a.k.a. $edges(KG)^+$. Alternatively, the set of *negative edges* of *KG*, a.k.a. $edges(KG)^-$, corresponds to $(V \times L \times V) - E$.

Open World Assumption. Under the Open World Assumption (OWA), *KG* can be incomplete, and some triples in $edges(KG)^-$ should be part of $edges(KG)^+$. Consequently, *KG* can be defined in terms of four types of edges: **i) Known Positive Facts:** correspond to the edges comprising *KG*, a.k.a. $kEdges(KG)^+$; **ii) Known Negative Facts:** represent the edges which are known to be false, a.k.a. $kEdges(KG)^-$; **iii) Unknown Positive Facts:** correspond to the edges in $edges(KG)^-$ which are true and should be part of $edges(KG)^+$; they are denoted by $ukEdges(KG)^+$; and **iv) Unknown Negative Facts:** are the edges in $edges(KG)^-$ which are false; they are denoted by $ukEdges(KG)^-$. Figure 1a illustrates a portion of the French Royalty *KG* [11]. The triple (dbo:Theobald_I_Navarre dbo:father dbo:Theobald_II_Navarre) is one of the unknown true facts in the *KG*, i.e., this triple belongs to $ukEdges(KG)^+$. The ideal knowledge graph KG^* , is the graph $(V, kEdges(KG)^+ \cup ukEdges(KG)^+, L)$.

Partial Completeness Assumption (PCA). The PCA heuristic assumes that negative edges (s, p, o') in $edges(KG)^-$ correspond to unknown positive edges in $ukEdges(KG)^+$, if there is an edge (s, p, o) in $edges(KG)^+$. This type of edges are called *heuristic-based negative edges* and are denoted by $hE(KG)^-$. Based on the PCA heuristic, unknown true facts— in Figure 1a— that correspond to the instantiations of the pattern (dbo:Theobald_I_Navarre dbo:father ?x) can be predicted following the PCA heuristic because the triple

(dbo:Theobald_I_Navarre dbo:father dbo:Theobald_III) already belongs to the *KG*. In general, the PCA heuristic can be applied to predict any triples (s, p, o') provided that a triple (s, p, o) exists within the *KG*, and the property *p* is not functional.

Inductive Knowledge. The acquisition of inductive knowledge necessitates the extraction of generalizable patterns from factual statements within a *KG*. This type of knowledge can manifest in either numeric or symbolic forms, and its acquisition can take on various methodologies such as supervised, self-supervised, or unsupervised, according to the chosen learning paradigm [10]. In self-supervised approaches, symbolic learning captures symbolic patterns inherent in a *KG*, e.g., expressed as Horn rules. Conversely, numerical learning generates numerical representations within a lower-dimensional embedding space. Unknown true facts can be inferred using either these embeddings or the symbolic rules, with scoring metrics like plausibility scores or standard confidence levels acting as indicators for the probability of an edge's veracity. They quantify the likelihood that a given edge pertains to $ukEdges(KG)^+$.

Horn Rules. A Horn rule is defined as follows: $Body \Rightarrow Head$. The body of the rule is comprised of predicate facts. The head is a predicate fact of a single atom. All the variables in the *Head* are terms of at least one predicate fact in the *Body*. At least one variable is shared by every two predicate facts in *Body* of the rule. A rule appears to be $R : B_1 \wedge B_2 \wedge B_3 \wedge \dots \wedge B_n \Rightarrow r(x, y)$, where *Head* corresponds to a binary predicate $r(x, y)$ that represents a triple pattern $(?x, r, ?y)$, and *Body* is the conjunction of binary predicates, $B_1 \wedge B_2 \wedge B_3 \wedge \dots \wedge B_n$, representing triple patterns.

Entailment of a Mined Rule. Given a directed edge-labeled graph $KG = (V, E, L)$ and a mined rule $r : Body \Rightarrow Head$, the entailed facts of *r* correspond to the instantiations of the predicate fact in *Head* on the substitutions of the variables in *Body* that are positive instantiations of the conjunction of the predicates in *Body*. That implies for all V_2 such that, $Body[Z := V_2]$ is a positive predicate fact of *Body*, $Head[Z := V_2]$ corresponds to an entailed fact of *r*. An entailed fact of $Head[Z := V_2] = p(s, o')$ of *r* is a positive entailed fact if $p(s, o') \in ukEdges(KG)^+$ a.k.a. *Unknown Positive Facts*.

Support of a Horn Rule. Given a directed edge-labeled graph $KG = (V, E, L)$ and a mined rule $r : Body \Rightarrow Head$, the support of *r* corresponds to the number of positive entailed facts of *Head*.

Standard Confidence of a Horn Rule Given a directed labeled-edge graph $KG = (V, E, L)$ and a mined rule $r : Body \Rightarrow Head$, the *Std_Confidence* of *r* corresponds to the ratio of the positive predicate facts of *Head* that are positive entailed facts based on *r*.

Partial Completeness Assumption Confidence of a Horn rule. Given a directed edge-labeled graph $KG = (V, E, L)$ and a mined rule $r : Body \Rightarrow Head$, where *Head* is $p(s, o)$. The *PCA_Confidence* score of *r* is the ratio of $support(r)$ to the cardinality of the union of $edges(KG)^+$ and $hE(KG)^-$. *PCA_Confidence* quantifies the amount of triples of the form $(s, p, ?x)$ from $edges(KG)^-$ that can be inferred following the PCA heuristic.

$$PCA(r) = \frac{support(r)}{|edges(KG)^+ \cup hE(KG)^-|}$$

Values equal to 1.0 or 0.0 indicate that no triples can be inferred.

Knowledge Graph Embedding Models. Numerical inductive learning models capable of transforming nodes and edges in a *KG*

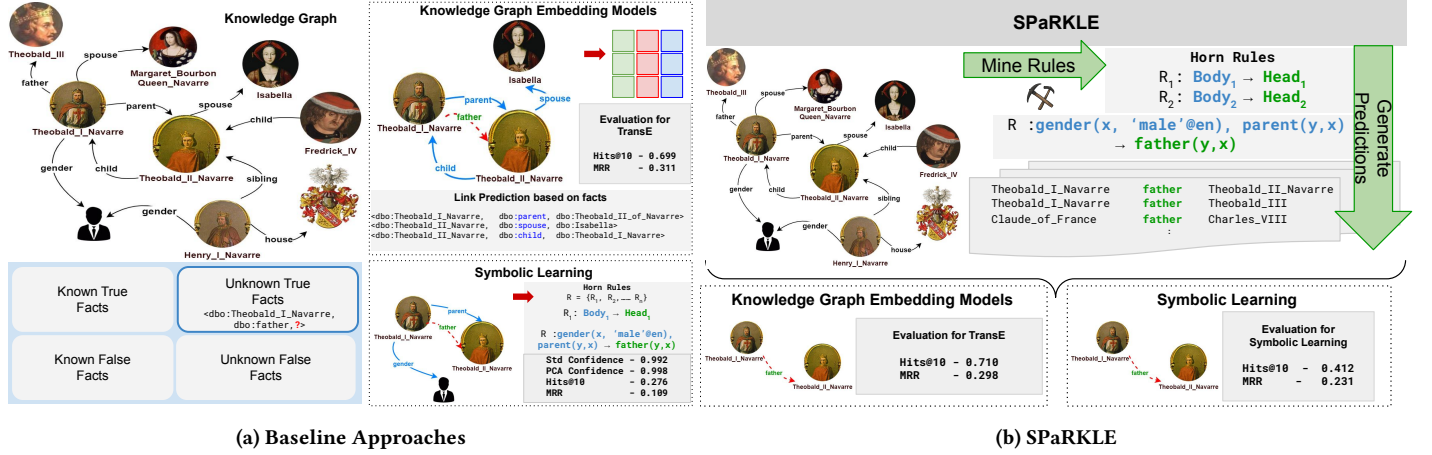


Figure 1: Motivation. Figure 1a depicts an exemplar sub-graph of the French Royalty KG. KGE models (e.g., TransE, TransD, ConvE, etc.) and symbolic learning techniques (e.g., AMIE or AnyBURL) are utilized to perform a prediction task of the missing link between two entities, i.e., Theobald_II_Navarre and Theobald_I_Navarre. Figure 1b shows *SPaRKLE*, in which the input KG is enriched with predictions produced from symbolic learning approaches to improve prediction performance to enrich KGs.

into a low-dimensional continuous vector space that preserves the KG’s structure. Given a directed edge-labeled graph $KG = (V, E, L)$ and a set of vectors T . A KGE of KG is a pair of mappings (ϵ, σ) such that i) $\epsilon: V \rightarrow T$, i.e., $\epsilon(e)$ maps an entity in V to a vector in T ; and ii) $\sigma: L \rightarrow T$, i.e., $\sigma(l)$ maps a directed edge l to a vector in T . A score function $\phi: V \times L \times V \rightarrow \mathbb{R}$ is used to measure the plausibility of the triples represented into latent vector space, triples $t = (s, p, o)$ with the higher score $\phi(s, p, o)$ values conveys better plausibility. The objective of KGE is to learn the embeddings in (ϵ, σ) that maximize the plausibility of positive edges in $edges(KG)^+$ and minimize the plausibility of negative edges in $edges(KG)^-$.

Hits@K. Given a tail prediction $(s, p, ?)$ over the directed edge-labeled graph $KG = (V, E, L)$, the model predicts a list of K entities that might be related to the tail entity. Hits@K determines the fraction of plausible entities that appear in the top K predictions.

$$Hits@K = \frac{Number of plausible entities \leq K}{K}$$

Mean Reciprocal Rank (MRR). Given the prediction task either from a head or tail perspective over the directed edge-labeled graph $KG = (V, E, L)$, the model ranks the plausible entities and calculates the reciprocal rank of the plausible entity for each predictive task.

$$MRR = \frac{1}{N} \sum_{i=1}^N \frac{1}{rank_i}$$

2.1 Motivating Example

Consider the exemplary sub-graph presented in Figure 1a, which focuses on the French Royalty KG, a repository of information about the French royal family. Within this KG, each member is connected through various relationships like *child*, *spouse*, *predecessors*, *gender*, *father*, and others. In the context of real-world KG representation, the OWA principle is employed. In this light, Figure 1a illustrates the presence of absent connections pertaining to a

specific entity, namely Theobald_II_Navarre. A practical scenario involves predicting these missing relationships. For instance, the prediction might entail establishing a link between two individuals, Theobald_II_Navarre and Theobald_I_Navarre, utilizing the father relation to signify the accurate RDF factual statement $\langle \text{Theobald_II_Navarre}, \text{father}, \text{Theobald_I_Navarre} \rangle$, which further contributes to the KG completion. Figure 1a illustrates the baseline approaches that work on the mechanism of numerical inductive learning and symbolic learning. The numerical inductive learning, i.e., KGE models predict the missing link via representing the entities and their relations in the low-dimensional vector space. For instance, predicting the tail entity, i.e., Theobald_I_Navarre, the KGE models (e.g., TransE) traverse the neighborhood of the head entity, i.e., Theobald_II_Navarre. Figure 1a states the metrics Hits@10 (0.699) and MRR (0.311) for TransE model. KGE models ignore the PCA heuristics that can be used to guide the link prediction process and create accurate predictions regarding missing relationships. Symbolic learning approaches, exemplified by systems like AMIE [12] (Association Rule Mining under Incomplete Evidence) and AnyBURL [15] (Anytime Bottom-Up Rule Learning), focus on inferring knowledge representation in the form of logical rules, specifically Horn rules of the form $r: Body \Rightarrow Head$. *SPaRKLE* currently relies on *AMIE* to generate predictions considering the PCA heuristics of the symbolic approach. An instantiation of a rule r is the process of substituting all the variables with entities from V . The head atom of an instantiated rule is called the predicted head atom if all body atoms of the instantiated rule exist in the input KG. During symbolic learning, various statistical measures are obtained to assess the quality of rules. According to [3], rules with maximum *Std_Confidence* is utilized to perform the LP task. Thus, in the exemplar sub-graph, the rule R (i.e., $gender(x, 'male'@en), parent(y,x) \Rightarrow father(y,x)$) with high *Std_Confidence* is selected to predict the missing link between two entities. The evaluation metric values for two AMIE predictions (Figure 1a) are 0.276 and 0.109, respectively. However, such learning algorithms are quantitatively capable of

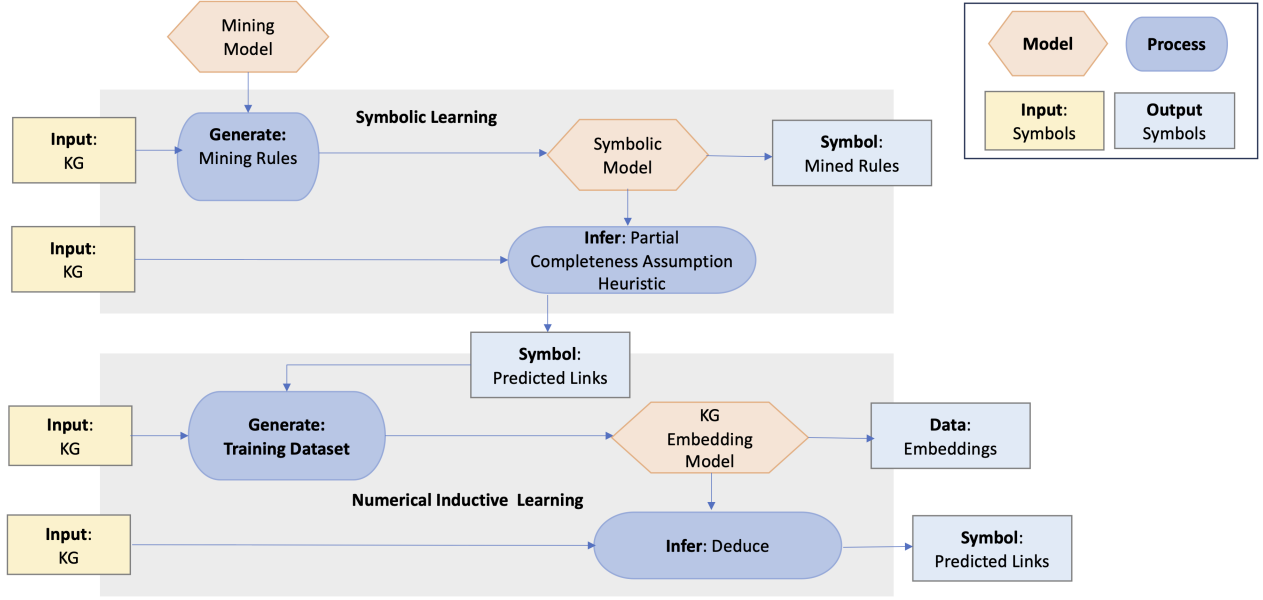


Figure 2: SPARKLE Design Pattern: Hybrid design pattern to demonstrate the usage of symbolic rules in combination with the sub-symbolic models. The symbolic learning component shows the execution of a symbolic model to infer the Partial Completeness Assumption (PCA) Heuristics also referred to as predicted links. These predicted links are further fed to the numerical inductive learning while training the KGE models to enhance the prediction of the models efficiently.

performing a prediction task but suffer from taking into account the sparse sub-graph (i.e., *less-connected*). *SPARKLE* is motivated by a hybrid approach, further, utilized to enhance the performance of both inductive and symbolic learning. Figure 1 highlights the impact of considering explicit patterns learned by symbolic techniques (AMIE) in the LP task. The obtained implicit information from symbolic learning is combined with the input KG, causing the metrics Hits@10 and MRR to improve, indicating an improvement in prediction performance. Figure 1b reports the improved performance of numerical techniques and symbolic learning techniques when implicit knowledge is provided with the original KGs. As a result, reported values for the KGE models are 0.710 for Hits@10 and 0.298 for MRR. On the other hand, the values for symbolic learning techniques also increase to 0.412 for Hits@10 and 0.231 for MRR. The outcomes of this study suggest that a hybrid approach *SPARKLE* that combines various techniques, such as symbolic learning and numerical learning, can significantly improve the quality and comprehensiveness of KG embeddings, particularly when dealing with entities with limited or sparse information. Therefore, *SPARKLE* can produce more accurate, and contextually rich embeddings with captured knowledge, i.e., $ukEdges(KG)^+$, to enrich KGs.

3 SPARKLE

The fusion of numerical inductive learning (data-driven) and symbolic approaches (knowledge-driven) stands as a significant challenge in contemporary AI. This section defines the problem of Link Prediction addressed in this paper and introduces our hybrid learning approach. Built on modular design principles for hybrid AI

systems as outlined in [20], we expound upon *SPARKLE* through the lens of a compositional design pattern, effectively integrating both symbolic and numerical learning components.

Problem Statement: Consider a knowledge graph $KG = (V, E, L)$, a subject entity s in V and predicate r in L , and a scoring function $\phi(s, r, o')$ quantifying the plausibility of a triple (s, r, o') . The problem of link prediction over $KG \langle s, r, ?x \rangle$ corresponds to the problem of identifying the entity o' in V that maximizes the value of $\phi(s, r, o')$. Formally, the problem of link prediction corresponds to the optimization problem of finding an entity o' that produces the most plausible answer for the incomplete triple $(s, r, ?x)$:

$$o' = \arg \min_{e \in V} \phi(s, r, e)$$

Proposed Solution: We propose a hybrid approach that synergizes symbolic and numerical learning techniques. The symbolic facet of our approach leverages the PCA heuristic to deduce missing triples through the application of rules acquired from symbolic learning processes. The plausibility assessment of a predicted triple by the symbolic system is anchored in the maximum value of the standard confidence observed in Horn rules that validate the projected triple. Our approach exclusively considers mined rules capable of generating triples with substantial plausibility. This selection process is guided by two defined thresholds: σ and θ . Specifically, only mined rules boasting a standard confidence surpassing σ , and a PCA confidence exceeding θ while remaining less than 1.0, are taken into account. The triples entailed by these selected mined Horn rules bolster the KG completeness, setting the stage for subsequent KGE

Table 1: Statistics about the benchmark KGs

Benchmarks	#triples	#entities	#relations
French Royalty KG	12,554	2,656	12
Family KG	28,358	3007	12
YAGO3-10	1,179,040	123,182	37

models to predict new missing links from a less sparse KG. *SPaRKLE* implements this hybrid approach; it facilitates the unification of symbolic learning models by generating potential heuristic edges that can then be fed into numerical models to improve the performance of predicting missing relationships in a KG. The modular design pattern in Figure 2 illustrates the usage of hybrid systems that incorporates symbolic learning to enrich the sub-symbolic learning; this *compositional* design pattern is devised based on the basic design patterns proposed by van Bekkum et al. [20] to describe hybrid AI systems. The design pattern comprises the inputs in the form of a KG, and produces three types of outputs: mined Horn clauses, predicted links and embeddings; inputs and outputs are represented with rectangular boxes. The system described with the pattern named *Symbolic Learning* receives a KG (yellow box) and based on mining model (depicted in pink hexagonal box) mined Horn clauses (shown in a blue rectangle). Links are predicted following the Partial Completeness Assumption (PCA) heuristic on the mined rules (blue oval rectangle). These predicted links represent prior knowledge for the KGE model; they are passed together with the KG to the system specified with the pattern *Numerical Inductive Learning*. This enhanced KG is used for generating a training dataset (shown in blue oval rectangle) which is utilized by the KGE model (depicted in pink hexagon) to learn the low-dimensional vector representation of the KG. The learned embeddings are used to deduce new links (shown in blue oval rectangle) based on a KG.

4 EXPERIMENTAL STUDY

We empirically evaluate the effectiveness of our approach at capturing knowledge and enriching the KG completion process. In particular, we aim to answer the following research questions: **RQ1**) Does the incorporation of implicitly acquired knowledge through symbolic learning significantly influence the quality of the Knowledge Graph Embedding (KGE) models? **RQ2**) To what extent do the Partial Completeness Assumption heuristics prove effective in enhancing a KG completeness? We set up the following experimental configuration to evaluate our research questions.

Benchmarks: We assess *SPaRKLE* on three established KGs: the French Royalty KG [9], the Family KG [11], and YAGO3-10 [14]. Table 1 depicts the statistics of all the benchmarks. The *French Royalty* KG is created by extracting information about French royal families from DBpedia, and for each person, we added the class *dbo:Person* as well as different relations like the number of children or predecessors, and further triple related counts. The *Family* KG describes relations between families. *YAGO3-10* is a fully curated subset of the YAGO3 that depicts a vast collection of entities and their relationships as factual statements that include a hierarchical taxonomy derived from Wordnet [16] and Wikipedia.

Baseline Models: We compare and utilize eight baselines for our approach *SPaRKLE*. The experiments used source codes of various

KGE models from the PyKEEN [5] pipeline, which includes TransE, TransH, TransD, ConvE, ComplEx, RotatE, and RESCAL. We use the same hyperparameter settings mentioned in the repository of PyKEEN except for the ComplEx model. We run the ablation study for tuning the hyperparameters for the ComplEx model concerning the benchmark KGs. The geometric models, such as TransE, TransD, and TransH translate the embedding space of the head entity with the given relation closer to the tail entity. Models like ComplEx, RESCAL, and RotatE learn the embedding representation in the complex vector space. RotatE is a popular model that has gained attention in learning the symmetric, asymmetric, one-to-many, and many-to-many relationships efficiently. ConvE is a deep convolutional neural network model that captures complex patterns and representations between entities and is capable of processing different types of relationships. The transformed embedding vector is then scored using the respective scoring function against all the tail entities to rank them based on their similarity to the predicted vector. Additionally, this ranking helps in identifying the most plausible tail entity for a given head entity and relation. Whereas, for symbolic learning, currently, *SPaRKLE* resorts to AMIE [12] for extracting and mining explicit patterns from the input KG. *SPaRKLE* is model agnostic and can be employed with any symbolic learning and numerical learning techniques. Table 2 highlights the comparison between baseline and *SPaRKLE* approaches of KG completion. **Metrics:** *Hits@K* quantifies the top-K ranking accuracy of anticipated links, counting how frequently the true (ground truth) links occur in the top-K ranked predictions provided by a model; we use Hits@10. Mean Reciprocal Rank (MRR) takes into account the first correct prediction in the list of rated forecasts, as well as the average of the reciprocal rank (i.e., $1/\text{rank}$). Both measurements have values between 0 and 1, and are *higher is better*. Additionally, *Execution Time* is measured as the elapsed time spent to execute the generation of the training dataset, learn the predictive model (i.e., mined rules or embeddings), and infer new links. The execution time is measured using the Python library `time`.

Implementation: *SPaRKLE* is implemented in Python 3.9 and executed in a virtual machine on Google Colab with 16 GiB VRAM and 1 GPU NVIDIA Tesla T4, with CUDA Version 12.0 (Driver 525.105.17) and PyTorch (v2.0.1). The implementation of our approach *SPaRKLE*, the benchmark KGs, and the trained KGE models are publicly available in our GitHub repository¹. Our implementation follows the hybrid design pattern as illustrated in Figure 2, strategically combining numerical inductive learning with the symbolic learning method. Symbolic learning is performed over the input KG, resulting in logical rules and PCA heuristic-based edges. The learned heuristic-based edges are used as prior knowledge to enhance numerical learning methods, i.e., KGE models. During symbolic learning, *SPaRKLE* utilizes extract logical rules with *PCA_Confidence* lower than 1 to infer heuristic-based negative edges, and the mined rules are applied to make predictions about missing links. These predictions are based on logical inference to compute the entailment of the mined rules. The process of *entailment of mined rules* to generate the heuristic-based negative edges (i.e., $hE(KG)^-$) is implemented using SPARQL queries over the portions of the KGs where the rules were learned. The thresholds σ and θ —for *Std_Confidence* and a *PCA_Confidence*—are set up 0.55 and 0.70, respectively. These values were tuned after several runs of the and supported by the

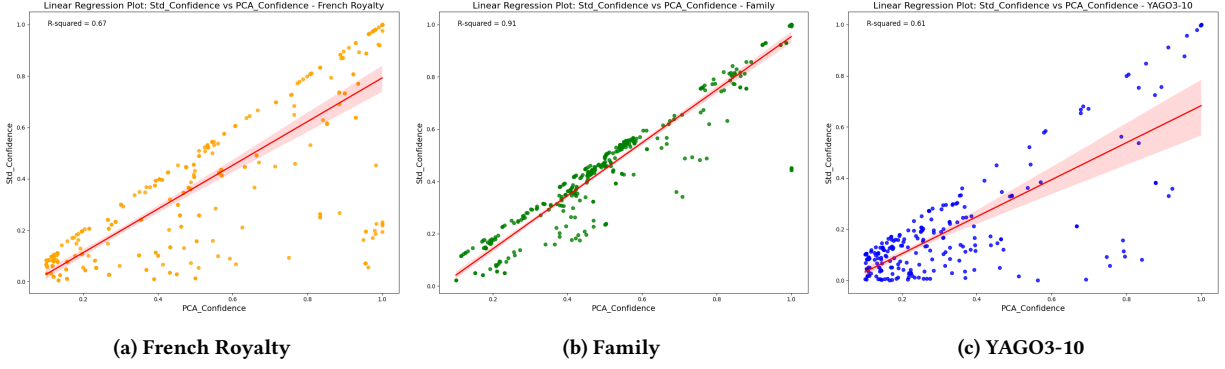


Figure 3: Linear Regression scatter plots with fitted regression lines to show the relationship between *PCA_Confidence* and *Std_Confidence* for KGs. A linear regression plot is a graphical depiction of the connection between two numerical variables.

results of a linear regression analysis reported in Figure 3. This analysis shows that there is a high correlation between *Std_Confidence* and *PCA_Confidence* in the rules extracted from the Family KG. As a result, the symbolic system may predict many more links of low accuracy, impacting, thus, the *SPaRKLE* performance.

The predictions generated from the symbolic learning system were then materialized in the input KG and fed as input to numerical inductive learning. We divided each KG into a 70 – 30 training-testing split. Our hybrid model learns to discern the KGE model’s behaviors during the training phase by processing the training data into low-dimensional space. This stage leverages the numerical learning component to capture intricate patterns in the training data. Here, the KGE models were optimized with the hyperparameters via running an ablation study with 5 trials. The model’s proficiency is assessed using Hits@K and MRR in the subsequent testing phase. In summary, *SPaRKLE* combines logical inference with data-driven insights, resulting in enhanced predictions captured from explicit and implicit relationships. *SPaRKLE* relies on [12] for symbolic learning. Moreover, our approach is model-agnostic and can incorporate other symbolic and numerical inductive learning methods.

4.1 Impact of the Partial Completeness Assumption Heuristic on Link Prediction

We report the empirical evaluation of the numerical inductive and symbolic learning approaches in terms of the evaluation metrics (i.e., Hits@K and MRR) proposed by Akrami et al. [3]. These empirical studies aim to answer the research questions **RQ1** and **RQ2**. We also present the statistical relationship of the rule mining metrics (i.e., *PCA_Confidence* and *Std_Confidence* Figure 3) in each of the studied KGs and their impacts on the outcomes of the KGE models. **SPaRKLE Effectiveness** is evaluated in terms of eight baseline approaches trained over existing KGs, i.e., $edges(KG)^+$, to predict missing links (see Table 2). Results of the KGE models reveal their performance in capturing global patterns not present in the KGs. On the other hand, symbolic logical rules $B \Rightarrow H$ capture such patterns and can be used to quantify the explanation for the predicted $ukEdges(KG)^+$. The results reported in Table 2 are obtained using the best hyperparameters provided by the PyKEEN optimizer².

²<https://pykeen.readthedocs.io/en/stable/reference/hpo.html>

We can observe that in the French Royalty KG (Table 2), the ComplEx model outperforms the other approaches with Hits@10 (0.855), while the RESCAL model performs the worst (0.226). Moreover, the performance of the RotatE model—in the Family KG—is the best (0.984). In comparison with the symbolic approach, in both KGs (i.e., the French Royalty and Family KGs), AMIE is capable of predicting the missing heuristic-based edges, and the reported values of Hits@10 and MRR are low compared to others. At the same time, the implementation of AMIE [12] uses no optimization strategies. In YAGO3-10, RotatE (0.439) and AMIE (0.384) are the best compared to others and align with the fact mentioned in [3] about AMIE capability in predicting $n - m$ relations. In contrast, the performance of TransE is significantly impacted, showing a decline of approximately 55.4%. These results corroborate findings by Akrami et al. [3] where AMIE exhibits the best predictive behavior. We can observe that *SPaRKLE* enhances the performance of all the models (with the average margin of 2.34%) as a result of incorporating the links entail from the AMIE mined rules into the input KG. The values reported in bold convey better results than the *Baseline* methods. For instance, considering the TransE model with the captured knowledge in the French Royalty and the Family KG, the performance is improved around 1.57% and 6.49%, respectively. Henceforth, the links predicted by *SPaRKLE* are more accurate, indicating that the captured implicit knowledge positively impacts the KGE models’ quality **RQ1**. Moreover, the values of AMIE (in the French Royalty KG and YAGO3-10) that were comparatively low (*Baseline*) are improved with a margin of 51.09%. Similarly, there is no substantial improvement when comparing the findings (particularly AMIE) of the Family KG. We also discovered an outlier within the KGE models, such as TransD values (in YAGO3-10), which are reduced by 0.84% (in terms of MRR). The rationale behind the low values is that the TransD model is susceptible to sparse sub-graphs. Figure 3 depicts the statistical relationship between *Std_Confidence* and *PCA_Confidence* scores for all the benchmark KGs. It is an effective statistical analysis for examining and quantifying the connection between two or more variables. It enables us to comprehend how changes in one variable are related to changes in another. Figure 3a reports on the distribution of the *PCA_Confidence* and *Std_Confidence* scores for the French Royalty KG; it demonstrates a moderate connection

Benchmark		French Royalty			Family			YAGO3-10		
Model	Approach	Hits@10	MRR	Time(s)	Hits@10	MRR	Time(s)	Hits@10	MRR	Time(s)
TransE	Baseline	0.699	0.311	36	0.679	0.201	61	0.194	0.072	6,414
	SPaRKLE	0.710	0.298	37	0.723	0.268	76	0.233	0.085	6,747
TransD	Baseline	0.551	0.164	59	0.708	0.224	102	0.239	0.108	17,400
	SPaRKLE	0.564	0.171	61	0.731	0.289	130	0.237	0.106	18,096
TransH	Baseline	0.819	0.515	57	0.655	0.224	103	0.169	0.088	10,425
	SPaRKLE	0.822	0.515	60	0.694	0.324	129	0.171	0.087	11,598
ConvE	Baseline	0.310	0.108	77	0.479	0.162	164	0.019	0.017	13,189
	SPaRKLE	0.382	0.140	87	0.565	0.208	209	0.020	0.018	16,906
ComplEx	Baseline	0.855	0.535	40	0.960	0.811	52	0.288	0.180	11,646
	SPaRKLE	0.875	0.516	43	0.973	0.806	54	0.362	0.241	11,988
RotatE	Baseline	0.764	0.491	39	0.982	0.782	67	0.439	0.244	11,332
	SPaRKLE	0.766	0.495	41	0.990	0.919	82	0.469	0.266	11,678
RESCAL	Baseline	0.226	0.050	66	0.012	0.008	133	0.003	0.001	9,027
	SPaRKLE	0.282	0.076	70	0.015	0.011	162	0.003	0.001	9,263
AMIE	Baseline	0.276	0.109	238	0.128	0.052	1,840	0.384	0.193	12,032
	SPaRKLE	0.471	0.309	289	0.124	0.054	3,338	0.427	0.236	13,717

Table 2: Empirical evaluation of various KGE models and symbolic learning technique (AMIE) over the different benchmark KGs. Hits@10 and MRR are reported. Light yellow color represents the baseline approaches. *SPaRKLE* (dark green color) indicates the impact of considering the captured knowledge in the prediction tasks. The values in bold depict better results.

between the *PCA_Confidence* and *Std_Confidence*. The data points of both the confidence scores tend to be closer to the regression line, showing that the model captures a good proportion of the variability in the *Std_Confidence*. The linear connection with the *PCA_Confidence* may explain approximately 67% of the variation in the *Std_Confidence*. A high R-squared value indicates that the independent and dependent variables have a strong linear relationship. This suggests that changes in *PCA_Confidence* are closely related to changes in *Std_Confidence*. Figure 3b demonstrates a high relationship between *PCA_Confidence* and *Std_Confidence*. The data points are grouped together tightly around the regression line, showing a strong linear connection. An R-squared score of 0.91 implies that the linear regression model accounts for about 91% of the variation in the *Std_Confidence*. Figure 3c shows that *PCA_Confidence* and *Std_Confidence* have an insignificant association. While the observed pattern states that the data points are more dispersed than in the preceding situation. The R-squared score of the linear regression model for YAGO3-10 is 0.61. These results indicate that in nearly 30% of the cases, mined rules exist within the French Royalty and YAGO3-10 KGs where the values of *Std_Confidence* are not dependent on the values of *PCA_Confidence*. Thus, there exist rules with lower values of *PCA_Confidence*, allowing the inference of links with higher *Std_Confidence*. This enables AMIE to perform significantly better in these two KGS than in the Family KG.

SPaRKLE Efficacy. The linear regression analysis allows us to determine the strength and relationship between *PCA_Confidence* and *Std_Confidence* of the mined rules. Predicting missing links in KGs can be significantly improved by following the PCA heuristic before applying an LP model. *SPaRKLE* relies on mined Horn rules representing patterns within a KG to generate contextually precise and semantically coherent predictions. This process contributes to the substantial enhancement of the KG’s overall completeness. In

direct comparison to existing methods [3], *SPaRKLE* not only confirms the efficacy of symbolic learning in predicting missing links but also distinguishes itself through its innovative employment of prediction tasks. This feature of the symbolic systems allows *SPaRKLE* to capture crucial patterns within densely interconnected sub-graphs more effectively. By incorporating the acquired insights into the input KG and subsequently retraining the baseline techniques, the performance of the LP models exhibits improvement as shown in Table 2. These results allow for answering **RQ1**. In the practical implementation of our *SPaRKLE*, we encountered the challenge of hyperparameter tuning for the KGE models, particularly the ComplEx model. To ensure reproducibility, our *SPaRKLE* implementation includes tuned hyperparameters optimized for the specific benchmark dataset. Further, Figure 3 reveals the effectiveness of enhancing the KG completeness using *SPaRKLE* for the benchmark KGs. This enables us to answer **RQ2**.

The SPaRKLE Efficiency is measured in terms of execution time—in seconds—for all the models. The observed overhead reveals that KGE models are highly computational. Particularly, all the models require two orders of magnitude more time for YAGO3-10 than for the other KGs (Table 1). However, we can observe that in YAGO3-10, the execution time of the prediction task over AMIE is faster (25.3%) compared to the ConvE model. Thus, we can conclude that combining the symbolic and numerical models requires higher execution time, but the overhead in execution time is negligible.

4.2 Discussion

SPaRKLE combines multiple approaches or components and can effectively deal with complex problems with few limitations. Despite *SPaRKLE*’s efficient performance, semantics is not considered during rule-mining over KGs. State-of-the-art mining-rule techniques rely on data graphs and often overlook the intrinsic semantics and

meaning of entities within KGs. It is essential to incorporate entailment regimes and deduction systems to enhance the accuracy of *SPARKLE* and exploit implicit facts and relationships. For instance, two entities (`dbp:Ned` and `dbp:Eddard`) connected with a relation `owl:sameAs`. Introducing entailment regimes allows for the inference of new implicit facts about `dbp:Eddard`, which are not present in the input KG. Moreover, this refinement leads to improved metric values, which, in turn, reveal the benefits of identifying true missing links or predictions with higher *PCA_Confidence* scores. *SPARKLE* is based on the PyKEEN pipeline, where the default batch size for training the model varies depending on the presence of a GPU. Without a GPU, the batch size defaults to 32, whereas with a GPU, it increases to 1024. Moreover, specific parameters, e.g., batch size, have an impact on the KGE models' performance. Generally, larger batch sizes tend to yield higher Hits@10 and Mean Reciprocal Rank (MRR) values. The results presented in Table 2 are achieved using GPUs, but this performance (i.e., accuracy and execution time) may differ when GPUs are not employed. Consequently, *SPARKLE* need to be empowered not only to deliver accurate results but also to provide sustainable solutions applicable to real-world scenarios. *SPARKLE* considers symbolic learning—through AMIE for mining logical rules— and sub-symbolic methods—utilizing KGE models. However, these approaches, and particularly the KGE models, lack inherent self-explanatory capabilities. For instance, when predicting the missing tail entity in a triple, `(dbp:Josh, dbp:nationality, ?)`, KGE models can successfully predict the missing link (`dbp:USA`) based on the known explicit facts about the head entity (`dbp:Josh`). Nevertheless, the latent vector representations used by KGE models may not explicitly reveal which explicit facts played the most crucial role in the link prediction task. As such, reasoning and explainable methods are required to empower *SPARKLE* with the capability of uncovering the facts in a KG that impact a given prediction.

5 RELATED WORK

The combination of symbolic and numerical strategies for enriching KGs is a viable option for improving the comprehensiveness and interpretability of these knowledge representation systems. Symbolic techniques thrive in capturing relationships based on rules, and logic-based dependencies among entities, but numerical approaches, rely on low dimensional vector space to identify connectedness between entities in huge KGs. Mostly KGE methods utilize triples from KGs as input and embeddings are trained using vector space assumptions [4], such as translational, neural network, and complex space assumptions. Further, the embeddings are obtained to perform the link prediction task [5]. Rivas et al. [18] propose a Neuro-symbolic perception for drug treatment response to enhance the link prediction capabilities of KGE models with the deduction of implicit knowledge using datalog rules. Akrami et al. [3] demonstrate the study conducted for a realistic and updated assessment of various KG completion techniques to establish their usefulness in improving KG completeness and quality. According to the findings of [3], the embedding models would have been biased toward learning reverse relations for LP due to data redundancy and Cartesian product relations present in the benchmarks. Furthermore, it was clearly demonstrated that simple models, such as symbolic learning approaches, outperform numerical models when

data contains reverse relations or data redundancy. As a result, we would like to showcase the combination of symbolic and numerical methodologies that can frequently result in enhanced performance and outcomes in a variety of activities, including KG completion in our proposed approach *SPARKLE*. While both symbolic and numerical techniques have advantages and weaknesses, combining them in a hybrid approach can mitigate shortcomings while leveraging the complementary advantages of these KG completion methods. In related research, Lajus et al. [12] introduce a symbolic learning technique that captures the co-occurrences of relationships, rules, and logical dependencies within KGs. Among various knowledge graph completion approaches, this method extracts association rules from KGs using the OWA (Open World Assumption). By deducing missing links and connections, *AMIE* significantly enhances both the quality and completeness of KGs, ensuring their augmentation while considering semantics encoded in KGs. Khajeh Nassiri et al. [17] pursue a symbolic learning approach that distinctly showcases the utilization of logical rules encompassing numerical predicates. This approach empowers KGs to understand and mine associations involving numerical values, measurements, and other quantitative attributes, culminating in a more expansive and precise representation of real-world knowledge. *SPARKLE* is designed to be system-agnostic, enabling seamless integration with any pre-existing KGE model or symbolic system. The prerequisite for integration is the computation of *PCA* and prediction scores.

6 CONCLUSIONS AND FUTURE WORK

SPARKLE is guided by the *PCA* heuristic, which improves predictions regarding missing KG links. Empirically, we have demonstrated that symbolic learning approaches can enhance—with prior knowledge—the performance of KGE models. Thus, *SPARKLE* exemplifies the benefits of integrating symbolic and numerical methodologies into a hybrid or neuro-symbolic AI system, allowing academics and practitioners to combine these two conceptual AI approaches, resulting in accurate solutions for KG completion. Albeit effective, hybrid approaches do come with limitations. This work shows that hybrid methods demand integrating various components, which can add computational complexity. Processing symbolic systems may result in the mining of rules and inference of triples unnecessary, which the numerical models will never utilize. Thus, this opens the door for future research to optimize computational complexity and avail the benefits of hybrid approaches. Secondly, hyperparameters related to different methods are frequently connected with hybrid models. The optimization of these parameters for the integrated model is essential. As a result, adjusting hyperparameters for both symbolic and numerical techniques will improve performance. Efficient hybrid models and automation methods for tuning them are part of our future research agenda. Moreover, frameworks like InterpretME [7] will be used for tracing the work of *SPARKLE* towards interpretable hybrid AI systems.

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